To What Extent Does Economic Activity Cause CO_2 Emissions?

Data Analysis 4 - Assignment 2

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1 Introduction

This report investigates the causal relationship between economic activity and carbon dioxide emissions using country-level panel data. The central question is: *to what extent does GDP per capita cause CO_2 emissions per capita?* A naive cross-sectional correlation between the two is large and highly significant, but it comines the causal effect with confounding factors, like geography, institutions, resource endowments, and industrial history that drive both variables. I use first-difference (FD) and fixed-effects (FE) panel methods to strip away these confounders and isolate the within-country, over-time relationship.

I estimate six models of increasing sophistication: two cross-sectional OLS regressions (for 2005 and 2023), three first-difference specifications (with 0, 2, and 6 lags of GDP growth), and a two-way fixed-effects model. I then add urbanization as a time-varying confounder to three of these models. The log-log functional form gives all coefficients a clean interpretation as *elasticities*: a 1% increase in GDP per capita is associated with a β% increase in CO_2 per capita.

2 Data and Variables

2.1 Source

All data come from the World Bank's *World Development Indicators* (WDI) database, covering 1992–2023. I downloaded four series:

- **GDP per capita, PPP** (constant international $): `NY.GDP.PCAP.PP.KD`
- **CO_2 emissions** (Mt, net GHG, AR5): `EN.GHG.CO2.MT.CE.AR5`

- **Population**: `SP.POP.TOTL`
- **Urban population (% of total)**: `SP.URB.TOTL.IN.ZS`

CO_2 per capita is constructed by dividing total CO_2 emissions (converted from megatonnes to tonnes) by population.

2.2 Functional form

Both GDP per capita and CO_2 per capita are heavily right-skewed in levels. The log transformation makes both distributions approximately symmetric (Figure 1). Moreover, the relationship between the two variables appears linear in log-log space (Figure 2), confirming that the log-log specification is appropriate. This is standard in the environmental economics literature and gives the regression coefficient an elasticity interpretation.

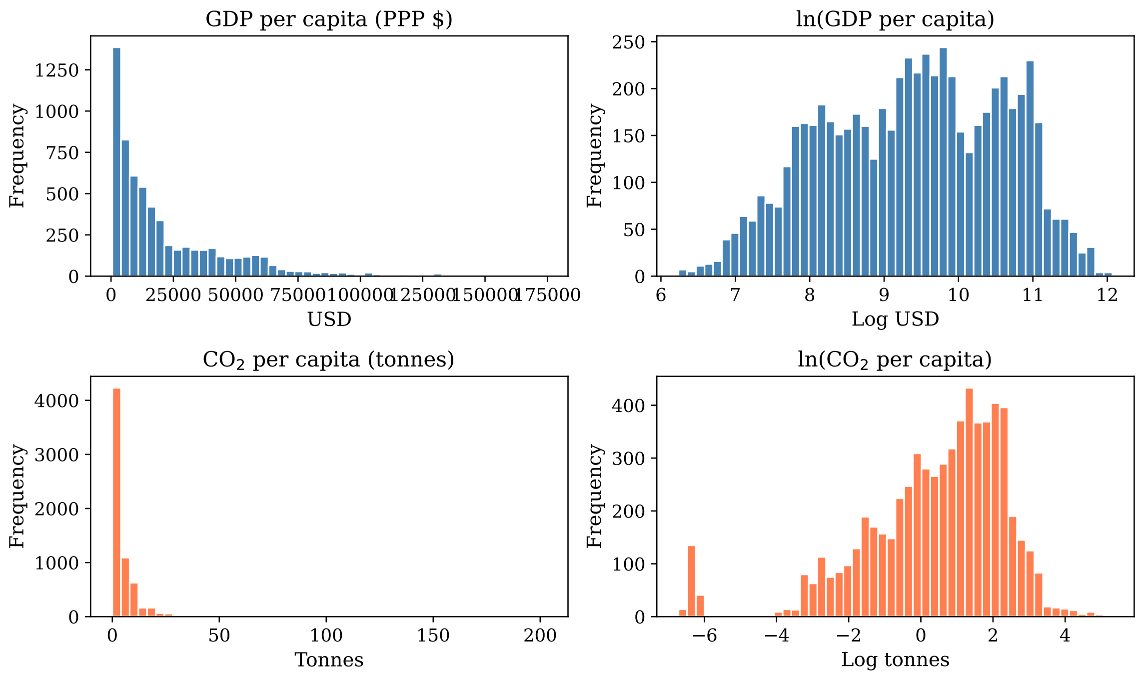

Figure 1: Distribution of GDP and CO_2 per capita in levels (left) and logs (right). The log transform removes the heavy right skew and produces approximately symmetric distributions.

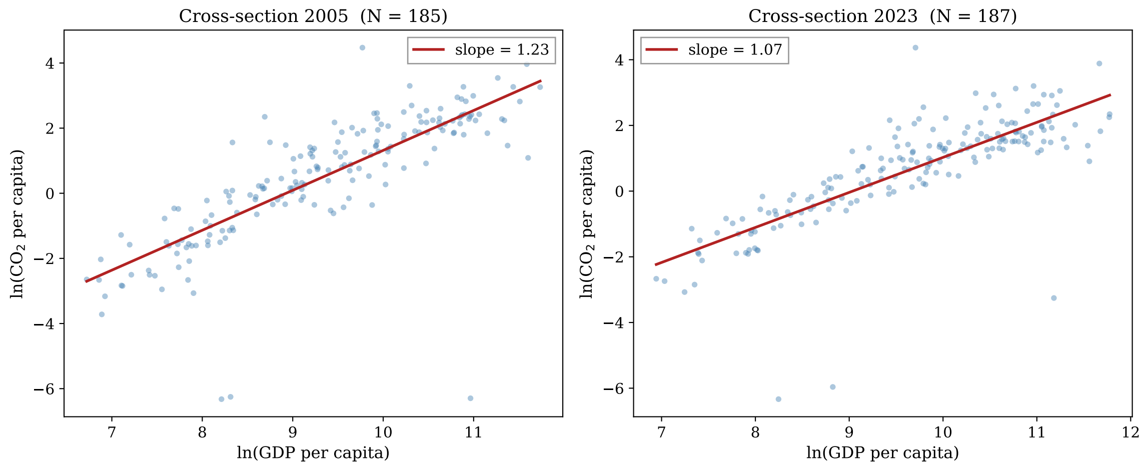

Figure 2: Cross-sectional relationship between ln(GDP pc) and ln(CO_2 pc) in 2005 and 2023. The relationship is approximately linear in log-log space with a stable slope ($\approx$ 1.0–1.2), confirming the log-log functional form.

3 Data Cleaning

3.1 Summary statistics

Table 1 reports summary statistics for the clean panel. GDP per capita ranges from under \$500 (Burundi, DRC) to over \$100,000 (Qatar, Luxembourg), reflecting the enormous cross-country income disparity. CO_2 per capita ranges from near zero (small island states, low-income African countries) to over 50 tonnes (Gulf states with large fossil fuel industries and small populations). Urbanization varies from under 10% to nearly 100%.

Table 1: Summary statistics — clean panel (190 countries, 1992–2023)

	N	Mean	Std. Dev.	Min	Median	Max
GDP per capita (PPP \$)	6,015	21,763	24,218	511	12,440	174,570
CO_2 per capita (t)	6,080	4.96	9.68	0.00	2.21	202.87
Urban population (%)	6,080	56.7	23.7	5.7	57.0	100.0
Population (millions)	6,080	35.0	133.2	0.01	6.5	1,438.1

3.2 Missing values

Figure 3 shows the missing data rate by year. Urban population and total population have zero missingness throughout the panel, as they are based on UN Population Division estimates. GDP per capita is the most incomplete variable ($\approx$10% missing), reflecting the difficulty of measuring economic output in fragile or small states. CO_2 data is somewhat more complete ($\approx$6.5% missing), though coverage deteriorates in the most recent years.

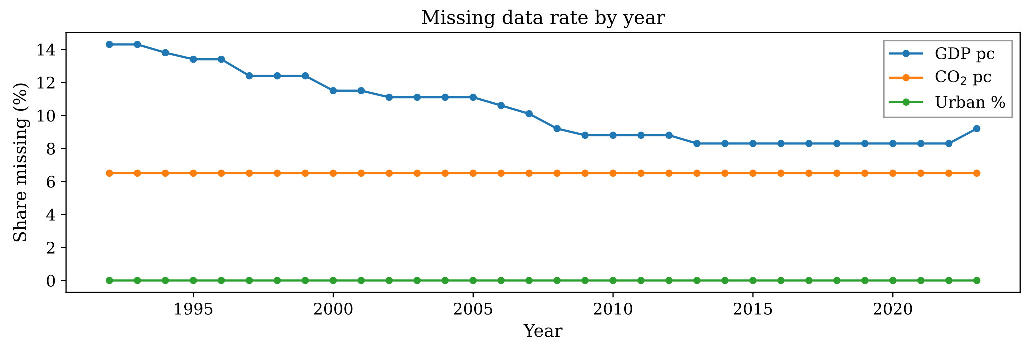

Figure 3: Share of missing observations by year for the three core variables. GDP per capita drives most of the data gaps.

3.3 Coverage and dropped countries

I assess coverage by counting, for each country, how many of the 32 years have all three core variables (GDP pc, CO_2 pc, urban %) non-missing. The distribution is bimodal (Figure 4): most countries have full or near-full coverage, but 27 countries have fewer than 16 complete years, including 26 with zero.

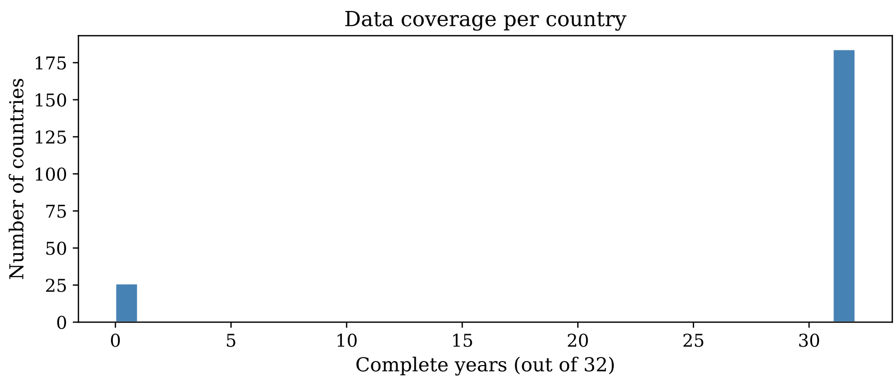

Figure 4: Number of complete years per country. The distribution is bimodal: most countries have full 32-year coverage, but 27 have fewer than half.

These 27 countries fall into clear categories:

- **Tiny territories** with no GDP data (American Samoa, Guam, Gibraltar, Channel Islands, Isle of Man, Liechtenstein, Monaco, San Marino, etc.)
- **Politically unstable or data-poor states** (North Korea, Cuba, Venezuela, Eritrea, Yemen, South Sudan, Kosovo, West Bank and Gaza)
- **Recently formed states** with no data before independence (Montenegro, Serbia, South Sudan)
- **Borderline case**: Djibouti (11 of 32 years)

Decision: I drop all 27 countries. These are either not meaningful economic units for the analysis (territories with populations under 100,000) or have data gaps too large for panel methods that rely on consecutive time-series observations. The clean panel contains **190 countries × 32 years = 6,080 observations**.

3.4 Outliers and zero emissions

I verify that extreme values are plausible. The highest CO_2 per capita observations belong to Qatar, Trinidad and Tobago, Kuwait, Bahrain, and the UAE. These are oil-rich Gulf states with small populations and massive fossil fuel industries. The lowest non-zero values are very low-income African countries (Burundi, DRC, Chad, Somalia) with minimal industrial activity. Both tails are genuine, not data errors.

A small number of country-year observations (64) have exactly zero CO_2 per capita, corresponding to tiny Pacific and Caribbean island states where reported emissions round to zero. These become missing under the log transform, which is the appropriate treatment as these countries contribute no meaningful variation for estimating the GDP–CO_2 elasticity.

4 Regression Analysis

I estimate six models, presented in Table 2. All models use $\ln(CO_2 \text{ pc})$ as the dependent variable and $\ln(\text{GDP pc})$ (or its first difference) as the main regressor. Standard errors are heteroskedasticity-robust (HC1) for OLS and clustered by country for all panel models.

Table 2: Main regression results

	OLS 2005 (1)	OLS 2023 (2)	FD (3)	FD 2 lags (4)	FD 6 lags (5)	FE (6)
ln(GDP pc)	1.226*** (0.075)	1.067*** (0.063)				0.609*** (0.079)
Δ ln(GDP pc)			0.431*** (0.058)	0.387*** (0.061)	0.408*** (0.069)	
Δ ln(GDP pc)$_{t-1}$				0.013 (0.052)	0.042 (0.060)	
Δ ln(GDP pc)$_{t-2}$				0.050 (0.030)	−0.004 (0.037)	
Δ ln(GDP pc)$_{t-3}$					0.049 (0.043)	
Δ ln(GDP pc)$_{t-4}$					0.067 (0.055)	
Δ ln(GDP pc)$_{t-5}$					−0.044 (0.052)	
Δ ln(GDP pc)$_{t-6}$					0.064 (0.042)	
Constant	−10.945*** (0.685)	−9.638*** (0.612)	−0.006 (0.009)	0.010 (0.009)	−0.005 (0.008)	
Country FE	No	No	No	No	No	Yes
Year dummies	No	No	Yes	Yes	Yes	Yes
Observations	185	186	5,763	5,387	4,635	5,951
R^2	0.629	0.624	0.055	0.054	0.059	0.966
R^2 Within						0.226

Notes: Dependent variable is ln(CO_2 pc) in columns (1)–(2) and (6), Δ ln(CO_2 pc) in columns (3)–(5). HC1 standard errors for OLS; clustered by country for panel models. $^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

4.1 Cross-section OLS (Models 1–2)

The OLS elasticity is 1.226 in 2005 and 1.067 in 2023, both highly significant ($p < 0.001$). These estimates exploit only *between-country* variation: richer countries emit more CO_2. The R^2 of ≈ 0.63 indicates that GDP alone explains nearly two-thirds of the cross-country variation in emissions.

However, these estimates are likely **biased upward**. Countries differ in many dimensions like geography, resource endowments, industrial structure, institutional quality that simultaneously determine both GDP and CO_2. The OLS coefficient captures the sum of the causal effect and all omitted variable bias. The slight decline from 1.23 to 1.07 over time may reflect a gradual decoupling of economic growth from emissions in richer countries.

4.2 First-difference models (Models 3–5)

First-differencing removes all time-invariant country characteristics, eliminating the largest source of omitted variable bias. The contemporaneous FD elasticity (Model 3) is **0.431**, roughly one-third of the OLS estimate. This means that within a country, a 1% increase in GDP in a given year is associated with a 0.43% increase in CO_2 *that same year*.

Adding lags (Models 4 and 5) allows us to capture the full dynamic response. The individual lag coefficients are mostly **individually insignificant**. This is expected and does not indicate that lags are unimportant. GDP growth rates are serially correlated, creating multicollinearity among the lag terms that inflates standard errors. What matters is the **cumulative (long-run) effect**:

Model	Cumulative elasticity
FD, no lags	0.431
FD, 2 lags	$0.387 + 0.013 + 0.050 \approx 0.450$
FD, 6 lags	$0.408 + 0.042 - 0.004 + 0.049 + 0.067 - 0.044 + 0.064 \approx 0.582$

The cumulative effect grows from 0.43 to 0.58 as more lags are included, suggesting that GDP shocks take several years to fully propagate into emissions through capital investment, infrastructure development, and changes in consumption patterns. The long-run FD elasticity of ≈ 0.58 is the preferred causal estimate from the FD family.

4.3 Fixed-effects model (Model 6)

The FE model with country and year fixed effects yields an elasticity of **0.609**, highly significant ($p < 0.001$). This is consistent with the long-run FD estimate, as expected: FE and long-run FD exploit similar within-country variation, but FE is more efficient because it uses both short- and medium-run variation without requiring explicit lag specification.

The within-R^2 of 0.226 indicates that GDP explains about 23% of the *within-country* variation in emissions after absorbing country and year fixed effects. This is substantially lower than the cross-sectional R^2 of 0.63, confirming that much of the cross-sectional association is driven by time-invariant differences between countries rather than by the causal effect of GDP.

5 Confounder Analysis: Urbanization

5.1 Why urbanization?

Urbanization (urban population as % of total) is a **common-cause confounder**: it independently affects both GDP per capita and CO_2 emissions. Urbanization drives industrialization and structural transformation of the economy, which increases GDP. Simultaneously, urban areas concentrate energy-intensive activities like transportation, construction, heating and cooling that increase CO_2 emissions. Crucially, urbanization is not simply a *mechanism* through which GDP affects CO_2 (which would make it a "bad control"). Rather, it is a structural characteristic that independently drives both variables.

In the panel context, urbanization is a **time-varying confounder**, it changes within countries over time, so it is *not* fully absorbed by country fixed effects. This makes it a valid and necessary addition to the FE and FD models, unlike time-invariant confounders (e.g., geography) that are already eliminated by the panel structure.

From the WDI database, urbanization data has near-complete coverage for all countries from 1992–2023, as it is based on UN Population Division estimates rather than survey data.

5.2 Results

Table 3 presents the confounder analysis, comparing each baseline model with and without urbanization.

Table 3: Confounder analysis: adding urbanization

	OLS (1)	OLS + U (2)	FD 2L (3)	FD 2L + U (4)	FE (5)	FE + U (6)
ln(GDP pc)	1.226***	1.196***			0.609***	0.584***
	(0.075)	(0.122)			(0.079)	(0.076)
Urban pop. (%)		0.002				0.016***
		(0.008)				(0.005)
Δ ln(GDP pc)			0.387***	0.390***		
			(0.061)	(0.060)		
Δ ln(GDP pc)$_{t-1}$			0.013	0.013		
			(0.052)	(0.052)		
Δ ln(GDP pc)$_{t-2}$			0.050	0.050		
			(0.030)	(0.030)		
Δ Urban (%)				−0.007		
				(0.005)		
Δ Urban (%)$_{t-1}$				0.003		
				(0.005)		
Δ Urban (%)$_{t-2}$				0.016**		
				(0.005)		
Country FE	No	No	No	No	Yes	Yes
Year dummies	No	No	Yes	Yes	Yes	Yes
Observations	185	185	5,387	5,387	5,951	5,951
R^2	0.629	0.630	0.054	0.057	0.966	0.966
R^2 Within					0.226	0.247

Notes: Odd columns: baseline models. Even columns: with urbanization as confounder. HC1 standard errors for OLS; clustered by country for panel models. $^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

OLS (columns 1–2): Adding urbanization barely changes the GDP coefficient (1.226 → 1.196), and urbanization itself is insignificant (0.002, $p > 0.1$). In the cross-section, GDP and urbanization are so highly correlated that the model cannot separate their effects.

FD 2 lags (columns 3–4): The GDP coefficients are virtually identical with and without the confounder (0.387 → 0.390). The contemporaneous urbanization change and its first lag are tiny and insignificant, but the **second lag is significant** (0.016, $p < 0.05$). This suggests that urbanization changes take approximately two years to affect emissions, a plausible delay reflecting the time needed for infrastructure construction and behavioral adaptation. The key finding is that the GDP coefficient is **unchanged**, indicating that urbanization does not confound the FD estimate.

FE (columns 5–6): Urbanization is **highly significant** (0.016, $p < 0.001$): a 1 percentage point increase in the urban share is associated with approximately 1.6% higher CO_2 emissions, controlling for GDP and year effects. The GDP coefficient decreases modestly from 0.609 to 0.584 (≈4% decline). This confirms that urbanization is a genuine time-varying confounder in the FE specification, but its inclusion does not substantially alter the main finding.

6 Summary

The naive cross-sectional association between GDP per capita and CO_2 emissions per capita is large (elasticity ≈ 1.1–1.2), but roughly half of this reflects omitted variable bias rather than a causal effect. Panel methods that exploit within-country variation over time (first-differencing and fixed effects) consistently yield a causal elasticity of approximately **0.4–0.6**. This means a 10% increase in GDP per capita causes roughly a 4–6% increase in CO_2 emissions per capita.

The FD models with lags reveal that the full effect takes several years to materialize: the contemporaneous elasticity is about 0.4, but the cumulative long-run effect reaches 0.58 when six lags are included. The FE estimate of 0.61 is consistent with this long-run effect.

Adding urbanization as a time-varying confounder confirms the robustness of these estimates. Urbanization has an independent, significant effect on emissions in the FE model, but its inclusion changes the GDP coefficient by only 4%. This is reassuring: the panel methods are already doing a good job of controlling for confounders, and the remaining time-varying confounders do not substantially bias the main estimate.

The overall conclusion is that economic growth does cause CO_2 emissions, but the causal effect is substantially smaller than the raw cross-country correlation suggests. A 1% increase in GDP per capita causes approximately 0.5–0.6% higher emissions, an elasticity well below unity, implying that CO_2 grows slower than GDP. This is consistent with the gradual "greening" of economic growth observed in many countries, though it also means that GDP growth alone, without active decarbonization policies, will continue to increase global emissions.